Non-Linear Control for a 4-DOF Robotic Manipulator

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ECEN 5738 Spring 2026

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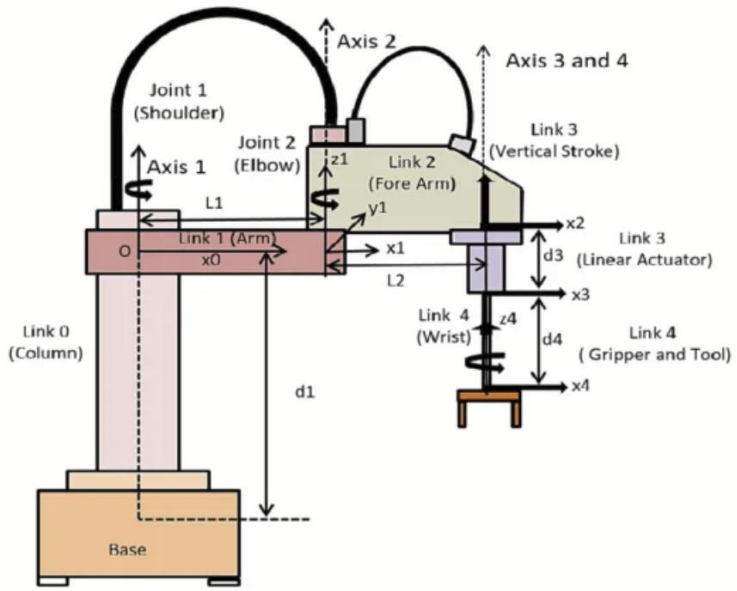
Agenda

- Problem statement
- Dynamics of SCARA Robot and Sliding mode control
- Literary review of papers discussing similar process
- System modeling in Matlab and Simulink
- Sliding mode vs. Backstepping vs. PID controllers



[10]

Problem Statement

- We modeled a 4 DoF SCARA robot with a Sliding mode controller and compared the results against a PID controller to analyze the error.

[11]

Dynamics

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + G(\theta) = \tau - p$$

- $\theta \in \mathbb{R}^{n\times 1}$ - joint space vector
- $M(\theta) \in \mathbb{R}^{n\times n}$ - positive definite inertia and mass matrix
  - **REASONING** $E_k = \frac{1}{2}\dot{\theta}^T M(\theta)\dot{\theta}$ and $E_k > 0 \; \forall \dot{\theta} \neq 0$
- $C(\theta, \dot{\theta}) \in \mathbb{R}^{n\times n}$ - centrifugal and Coriolis matrix
- $G(\theta) \in \mathbb{R}^{n\times 1}$ - gravitational vector
- $\tau \in \mathbb{R}^{n\times 1}$ - input torque vector
- $p \in \mathbb{R}^{n\times 1}$ - perturbations vector

Dynamics

$$M(\theta)\ddot{\theta} + C(\theta,\dot{\theta})\dot{\theta} + G(\theta) = \tau - p$$

$$D(\theta,\dot{\theta},\ddot{\theta}) = M_u(\theta)\ddot{\theta} + C_u(\theta,\dot{\theta})\dot{\theta} + G_u(\theta) - p$$

Isolate Disturbances and unknowns

$$M_k(\theta)\ddot{\theta} + C_k(\theta,\dot{\theta})\dot{\theta} + G_k(\theta) + D(\theta,\dot{\theta},\ddot{\theta}) = \tau$$

Plug back into dynamics

$$\ddot{\theta} = M_k^{-1}(\theta)[\tau - C_k(\theta,\dot{\theta})\dot{\theta} - G_k(\theta)] - M_k^{-1}(\theta)D(\theta,\dot{\theta},\ddot{\theta})$$

Isolate acceleration

Dynamics

$$\ddot{\theta} = M_k^{-1}(\theta)[\tau - C_k(\theta,\dot{\theta})\dot{\theta} - G_k(\theta)] - M_k^{-1}(\theta)D(\theta,\dot{\theta},\ddot{\theta})$$

GOAL: Design input τ so that the known nonlinearities are eliminated and there is some remaining virtual control: v

$$\tau = C_k(\theta,\dot{\theta})\dot{\theta} + G_k(\theta) + M_k(\theta)v$$

$$\ddot{\theta} = v + M_k^{-1}D(\theta,\dot{\theta},\ddot{\theta})$$

Dynamics

$$\ddot{\theta} = v + M_k^{-1} D(\theta, \dot{\theta}, \ddot{\theta})$$

Bounding Unknowns and Disturbances

$$w = M_k^{-1} D(\theta, \dot{\theta}, \ddot{\theta})$$

ASSUME: w can be bounded by some $||w_{max}||$

REASONING: System has physical limits and could incorporate safety barrier functions

$$\ddot{\theta} = v + w$$

State Space Equation

$$x_1 = \theta$$

$$x_2 = \dot{\theta}$$

$$\dot{x}_1 = \dot{\theta} = x_2$$

$$\dot{x}_2 = \ddot{\theta} = v + w$$

Sliding Mode Control

Define a new surface S:

$$S = \dot{E} + \Lambda E$$

$$\dot{S} = \ddot{E} + \Lambda \dot{E}$$

- $E = \theta - \theta_D = x_1 - x_{1D}$
- $\dot{E} = \dot{\theta} - \dot{\theta}_D = x_2 - x_{2D}$
- Λ - some diagonal gain matrix

Sliding Mode Control

$$v = \ddot{x}_D - \Lambda \dot{E} - K sgn(S)$$

- $K \in \mathbb{R}^{n \times 1}$ - positive definite diagonal matrix
- $\lambda_{K_i} > \lambda_{w_i}$

Final Dynamics With Sliding Mode Control

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \dot{x}_{2D} - \Lambda \dot{E} - K sgn(S) - w$$

Dynamics

$$M(\theta)\ddot{\theta} + C(\theta,\dot{\theta})\dot{\theta} + G(\theta) = \tau - p$$

- $\theta \in \mathbb{R}^{n\times 1}$ - joint space vector
- $M(\theta) \in \mathbb{R}^{n\times n}$ - positive definite inertia and mass matrix
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- $\tau \in \mathbb{R}^{n\times 1}$ - input torque vector
- $p \in \mathbb{R}^{n\times 1}$ - perturbations vector

Bounding Unknowns and Disturbances

$$D(\theta,\dot{\theta},\ddot{\theta}) = M_u(\theta)\ddot{\theta} + C_u(\theta,\dot{\theta})\dot{\theta} + G_u(\theta) - p$$

$$M_k(\theta)\ddot{\theta} + C_k(\theta,\dot{\theta})\dot{\theta} + G_k(\theta) + D(\theta,\dot{\theta},\ddot{\theta}) = \tau$$

$$\ddot{\theta} = M_k^{-1}(\theta)[\tau - C_k(\theta,\dot{\theta})\dot{\theta} - G_k(\theta)] - M_k^{-1}(\theta)D(\theta,\dot{\theta},\ddot{\theta})$$

GOAL: Design input τ so that the known nonlinearities are eliminated and there is some remaining virtual control: v

$$\tau = C_k(\theta,\dot{\theta})\dot{\theta} + G_k(\theta) + M_k(\theta)v$$

ASSUME: w can be bounded by some $||w_{max}||$

$$w = M_k^{-1}D(\theta,\dot{\theta},\ddot{\theta})$$

STATE SPACE EQUATION:

$$x_1 = \theta$$

$$x_2 = \dot{\theta}$$

$$\dot{x}_1 = \dot{\theta} = x_2$$

$$\dot{x}_2 = \ddot{\theta} = v + w$$

Don't exactly know w. Introduce some error term: E which is deviation from desired trajectory (due to w)

Sliding Mode Control: Define a new surface S:

$$S = \dot{E} + \Lambda E$$

$$\dot{S} = \ddot{E} + \Lambda \dot{E}$$

- $E = \theta - \theta_D = x_1 - x_{1D}$
- $\dot{E} = \dot{\theta} - \dot{\theta}_D = x_2 - x_{2D}$
- Λ - some diagonal gain matrix

Implementing a control law:

$$v = \ddot{x}_D - \Lambda\dot{E} - K sgn(S)$$

- $K \in \mathbb{R}^{n\times 1}$ - positive definite diagonal matrix
- $\lambda_{K_i} > \lambda_{w_i}$

Lyapunov Function Proof

$$V = \frac{1}{2} S^T S \tag{1}$$

$$\dot{V} = S^T \dot{S} \tag{2}$$

$$\dot{V} = S^T(\ddot{E} + \Lambda \dot{E}) = S^T(\dot{x}_2 - \dot{x}_{2D} + \Lambda \dot{E}) \tag{3}$$

$$\dot{V} = S^T[\dot{x}_{2D} - \Lambda \dot{E} - K sgn(S) - w + \dot{x}_{2D} + \Lambda \dot{E}] \tag{4}$$

$$\dot{V} = S^T[-K sgn(S) - w] \tag{5}$$

$$\dot{V} \leq -\sum_{i=1}^{4} |s_i|(|k_i sgn(s_i)| + |w_i|) \tag{6}$$

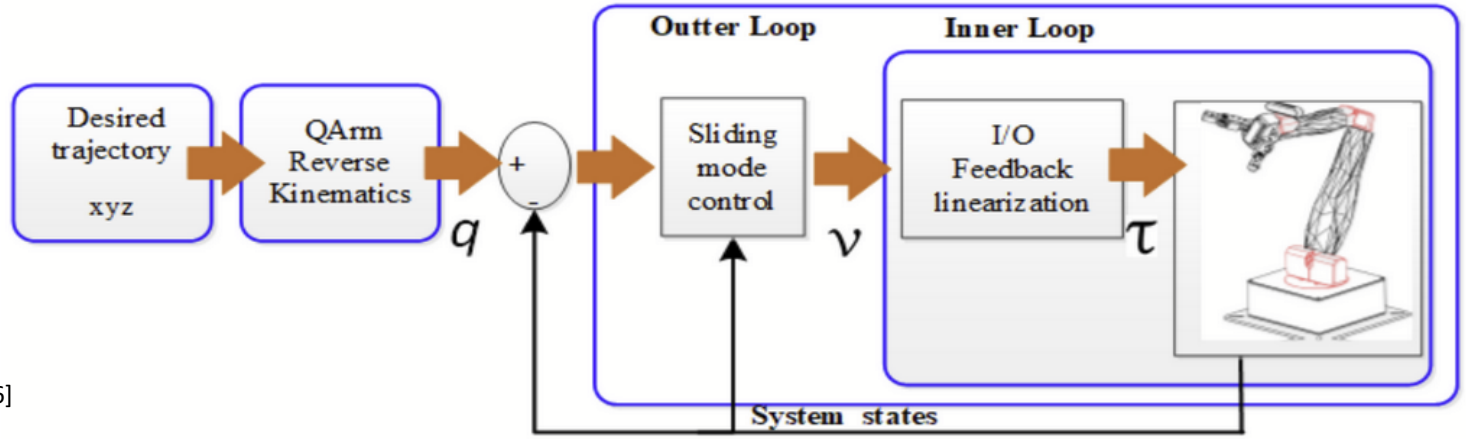
$$\dot{V} \leq -\sum_{i=1}^{4} |s_i|(|k_i| - |w_i|) \tag{7}$$

Nonlinear control design research

- Researched papers describing way for designing nonlinear controllers for 4 DoF robots.
- Majority used multiple advanced control techniques.
- Common comparison method was against a PID controller comparing error for each and showing large improvement with the more advances control techniques.

Four DOF Robot Manipulator Control Using Feedback Linearization Based on Sliding Mode Control

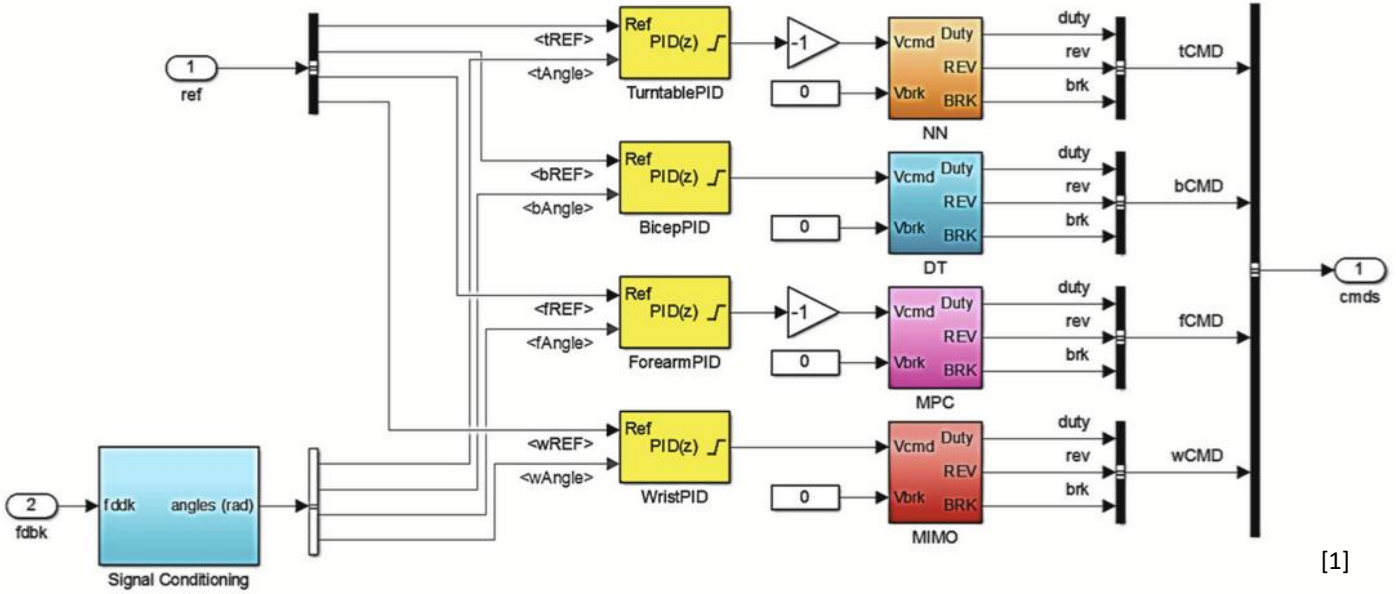
- Nonlinear control of Quanser Qarm
- Implemented Sliding Mode Control
- Error was compared against PID controller for the same 4 DoF robot resulting in over 78% error reduction [6]



[6]

Advanced multi-loop control for 4DOF robotic arms: Integrating digital twins, neural networks, and model predictive control

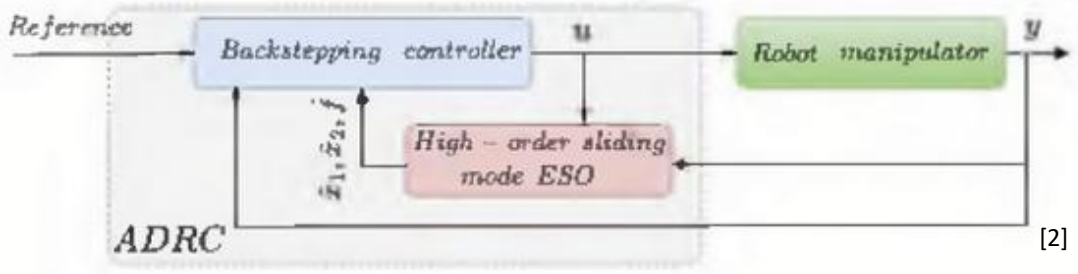
- 4 DoF robotic arm control using multiloop control, digital twin, Neural Networks (NN) and Model Predictive Control (MPC)
- MPC used a Nonlinear Model vs a Linear model for better state estimation.
- This control system reduced overshoot by 18%, settling time by 23% and increased tracking accuracy by 15%. [1]



[1]

Backstepping Active Disturbance Rejection Control for a 4 DoF Robot Manipulator Using High-Order Sliding Mode Extended State Observer

- 4 DoF robot controller modeled using disturbance rejcection including high order sliding mode control in conjunction with back stepping control.
- Extended state is used to account for disturbance by adding another state to the dynamics. (expand on this or remove)
- Compared results against only backstepping control and showed significant improvement of error tracking. [2]



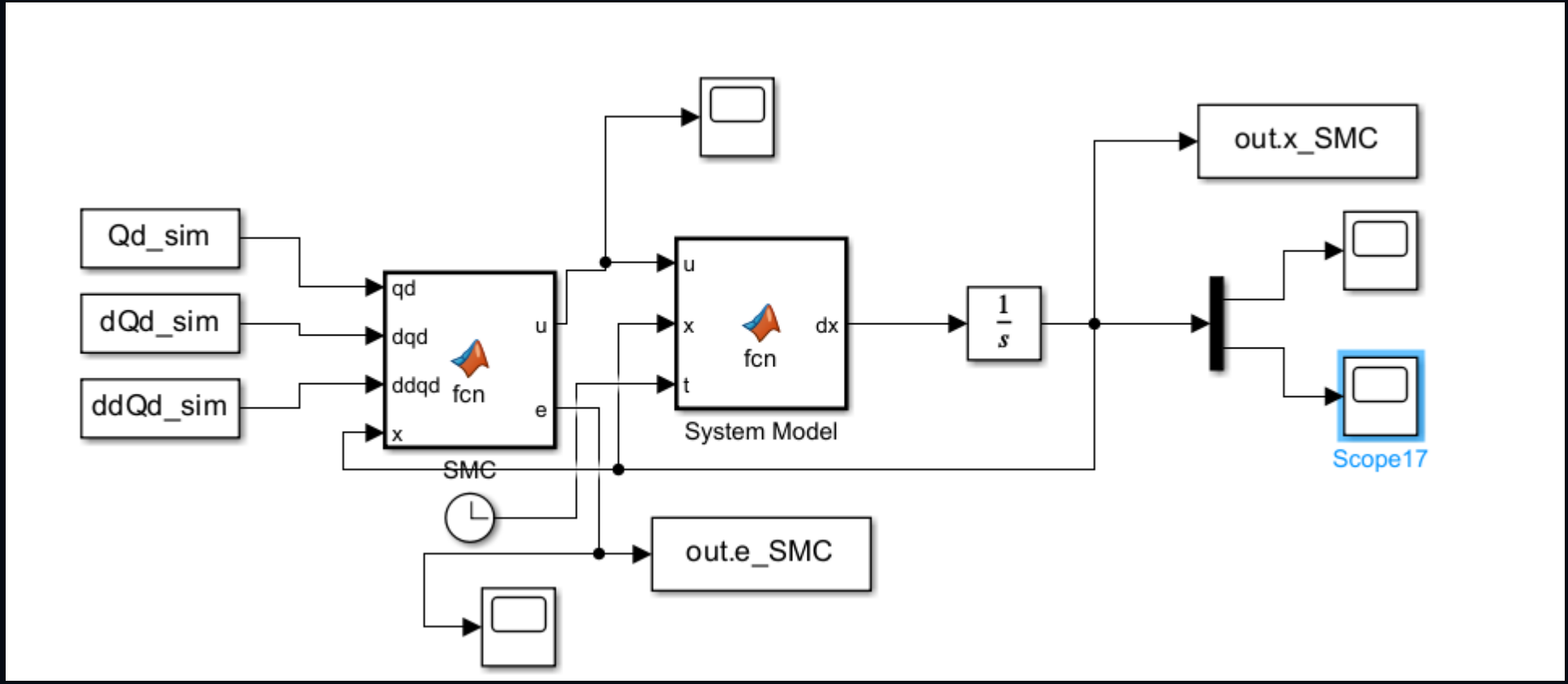
[2]

Controller design procedure

- Modeling 4 DoF robot in Simulink using dynamics of SCARA robot
- Design PID controller for simulation comparison
- Design sliding mode controller and backstepping controller for the SCARA robot

Simulations

Simulink SMC Block Diagram



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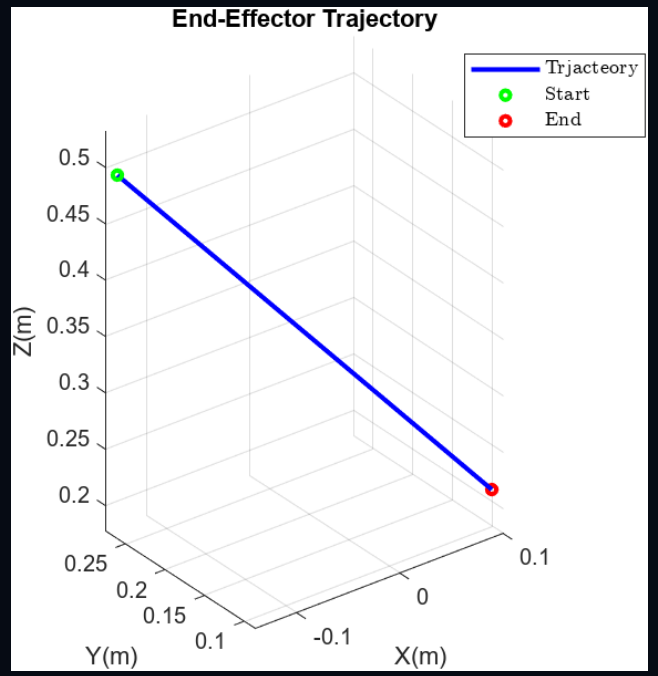
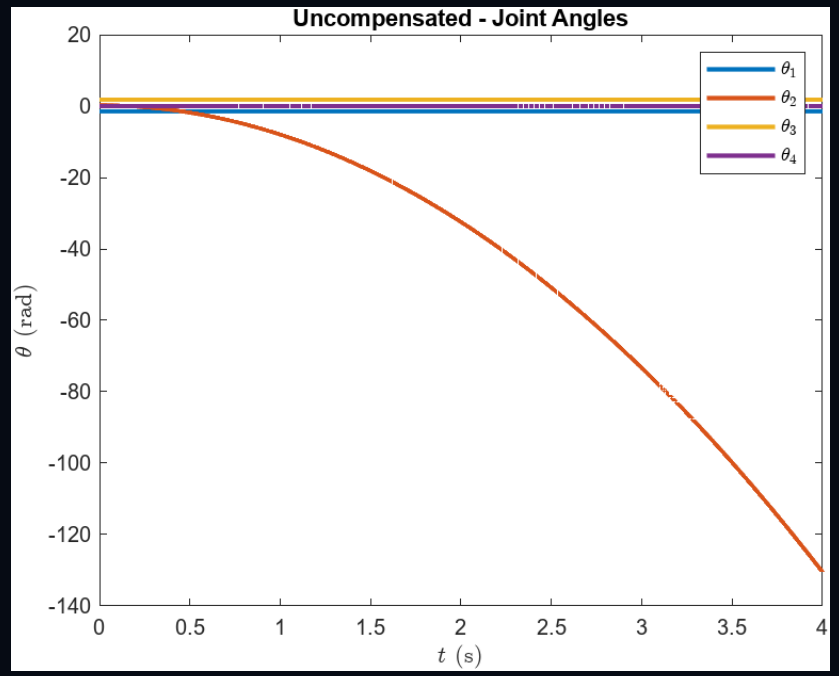
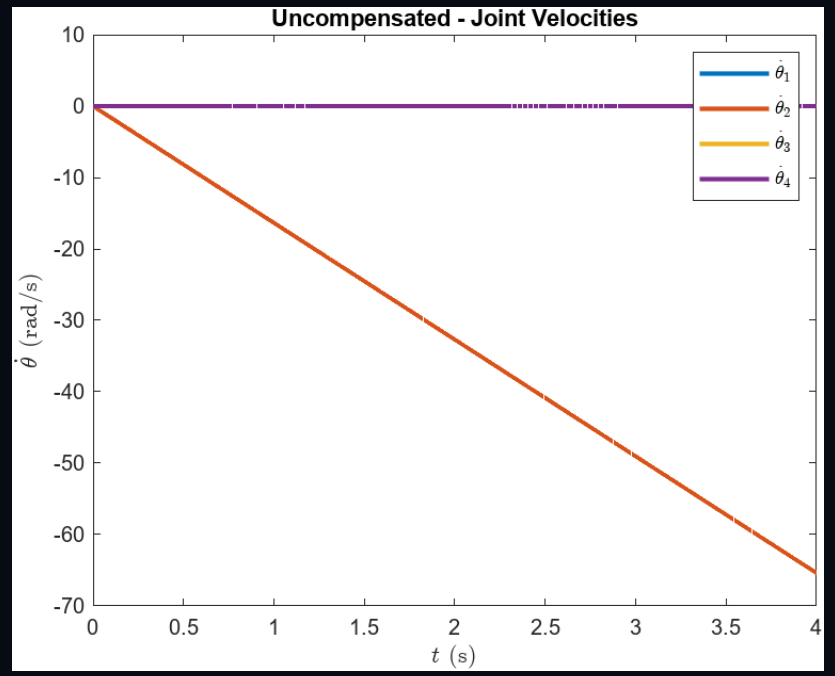
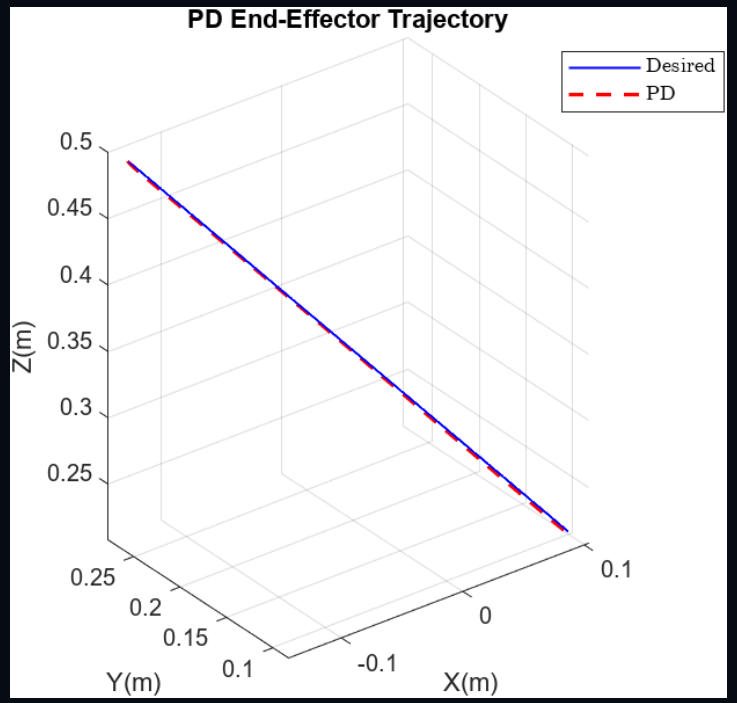
Uncompensated System

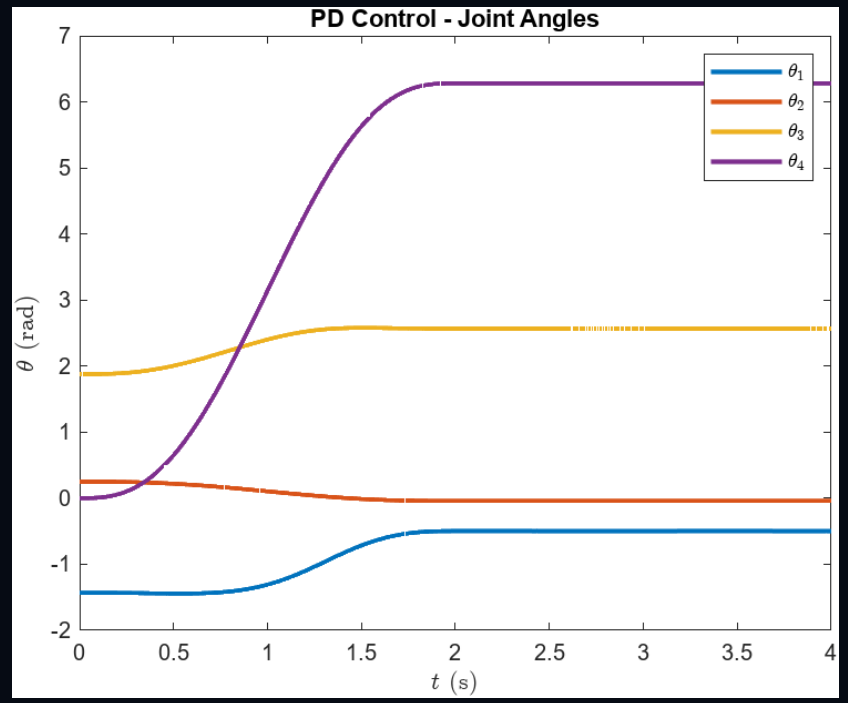


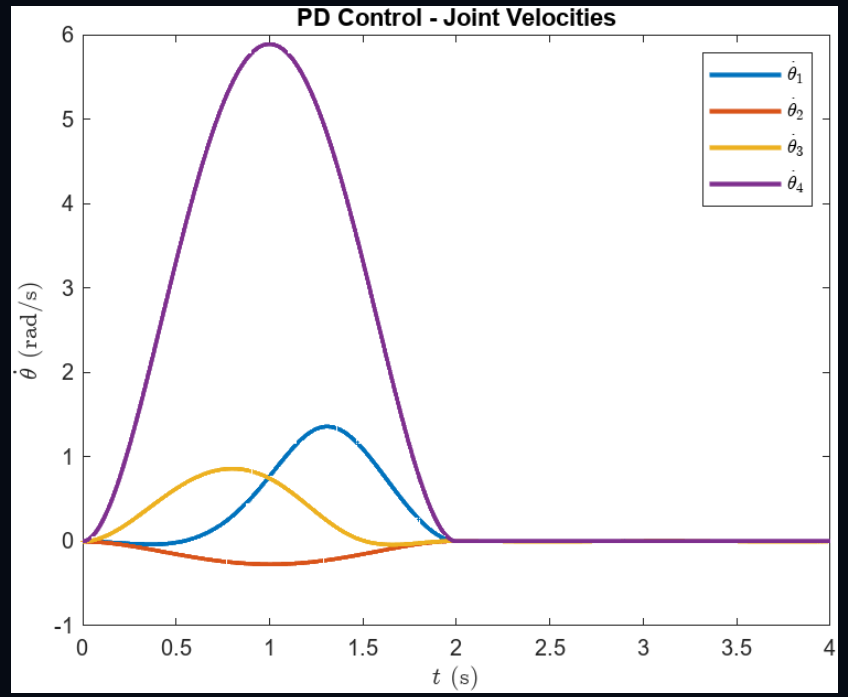




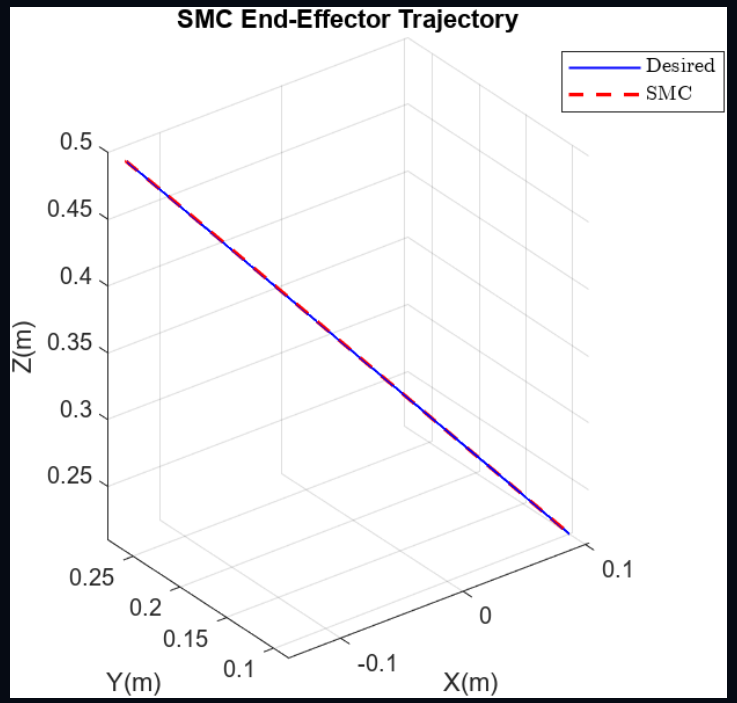
PD Controller

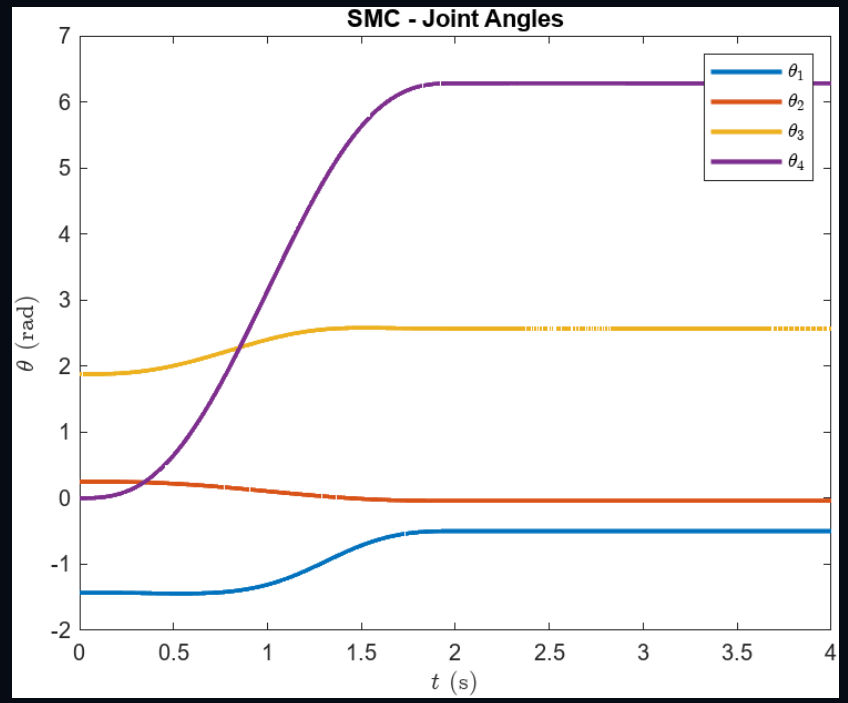


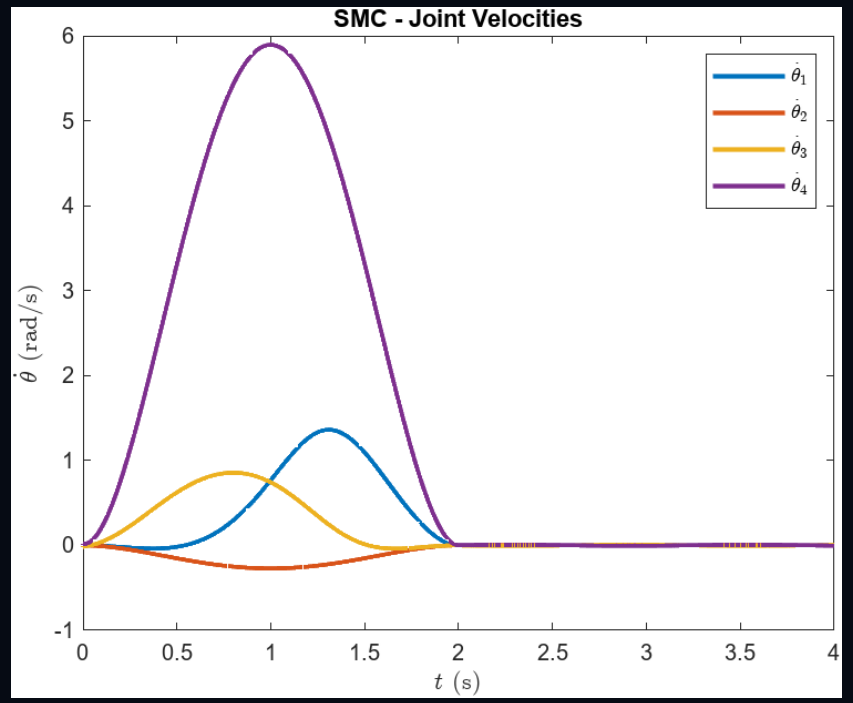




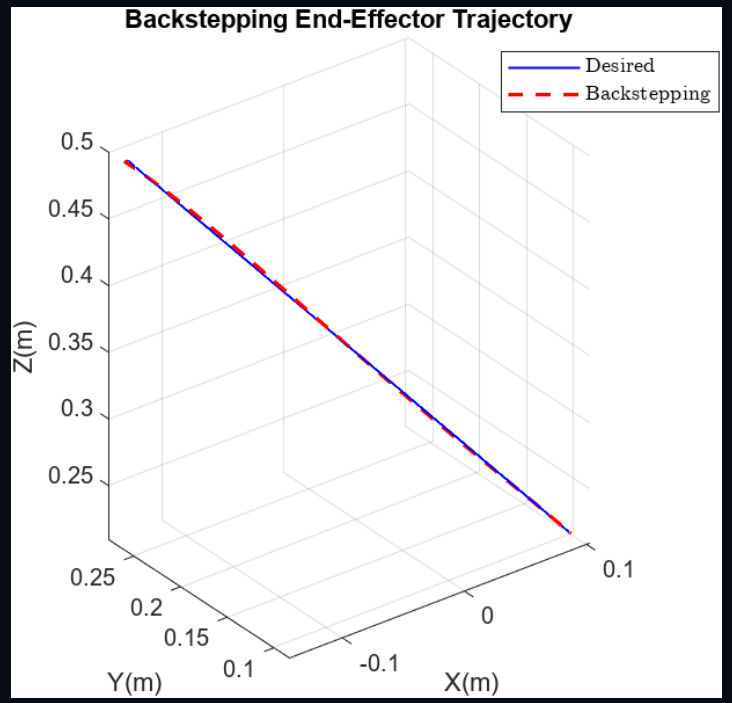
Sliding Mode Controller

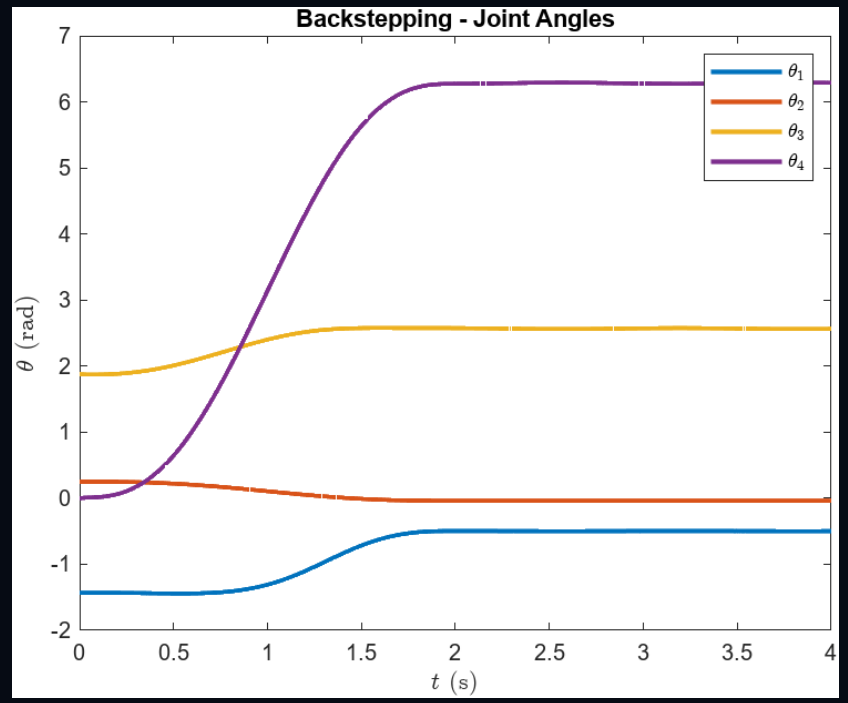


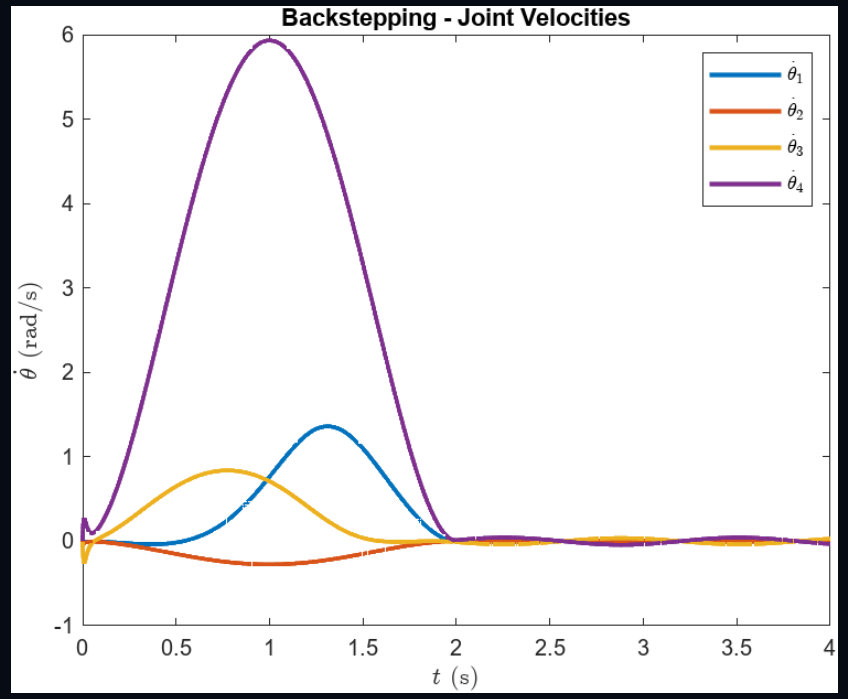




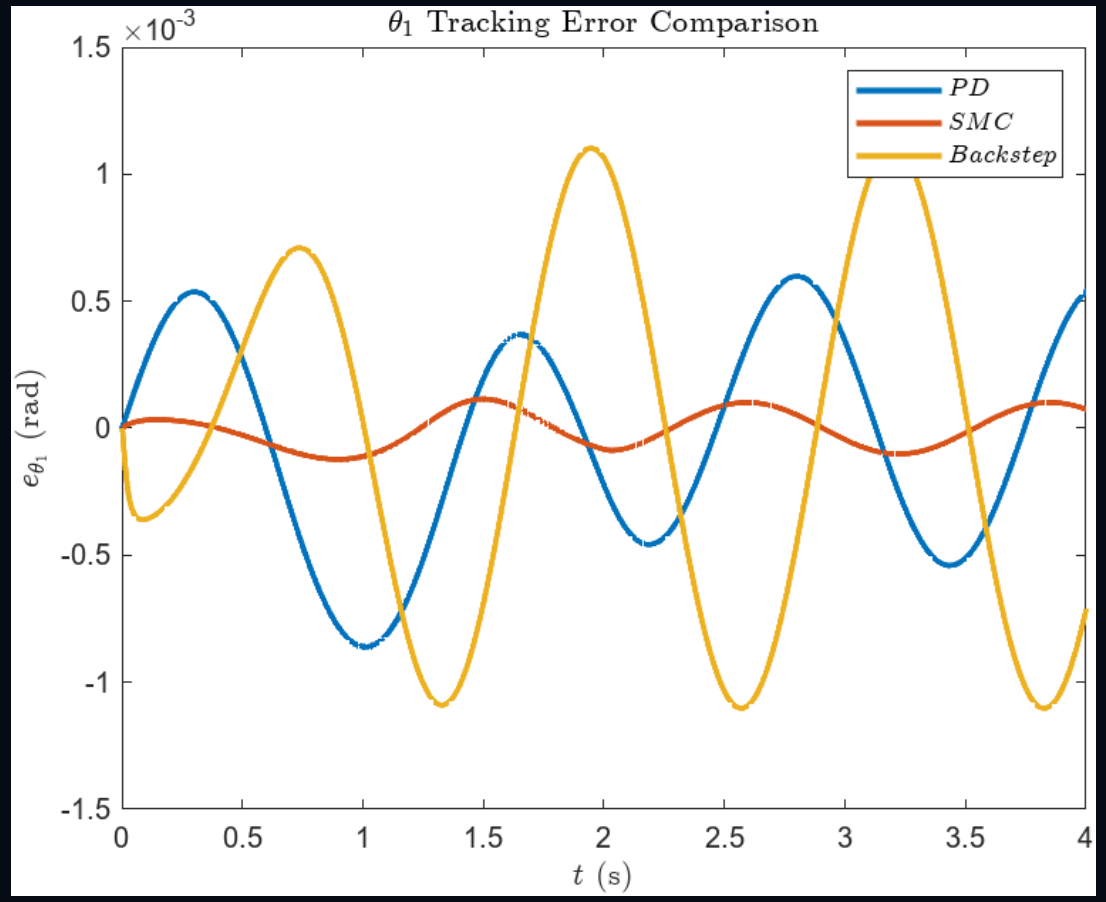
Backstepping Controller

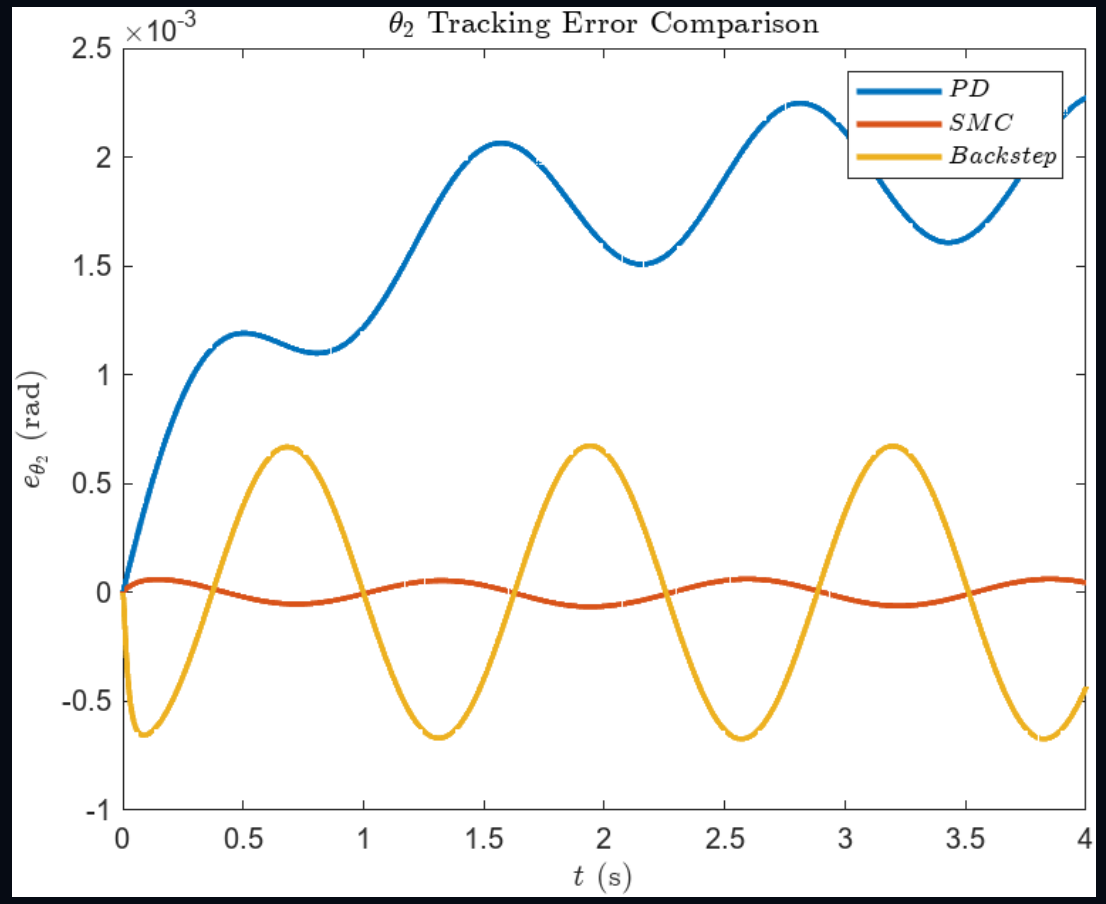






Tracking Error Comparison





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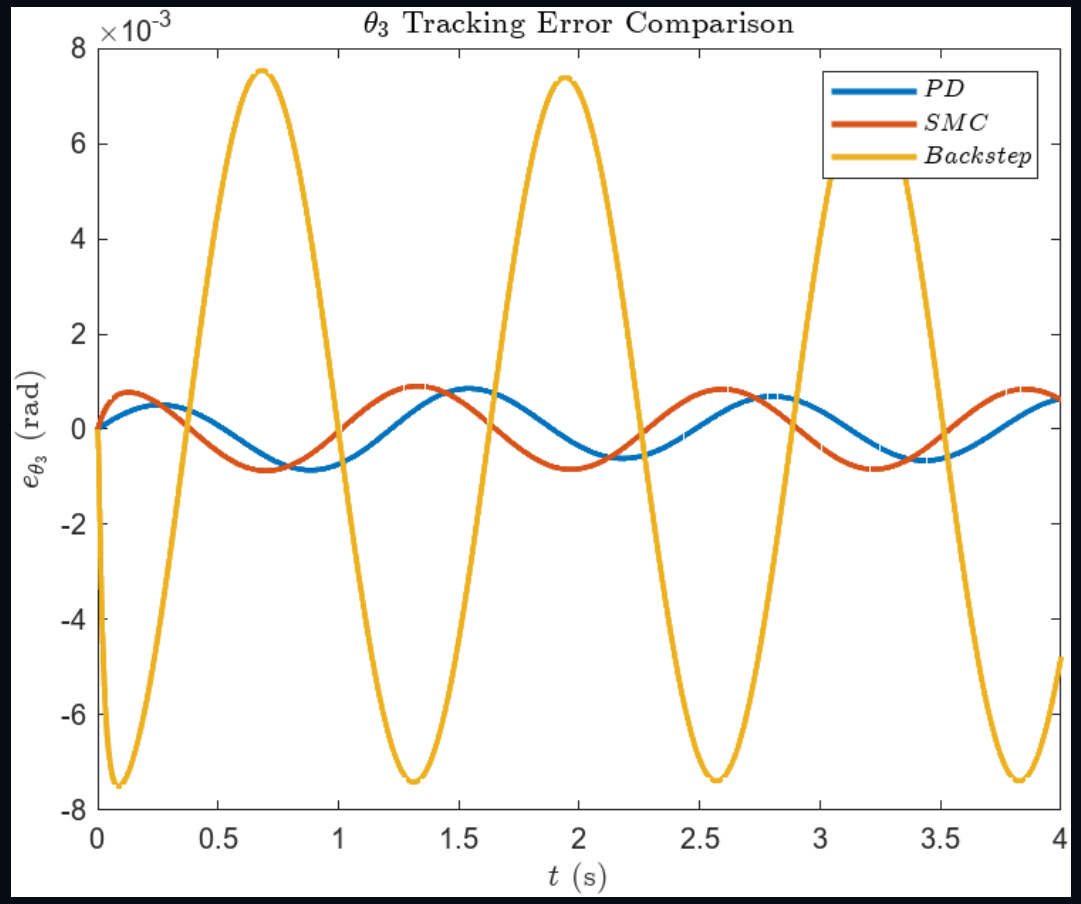
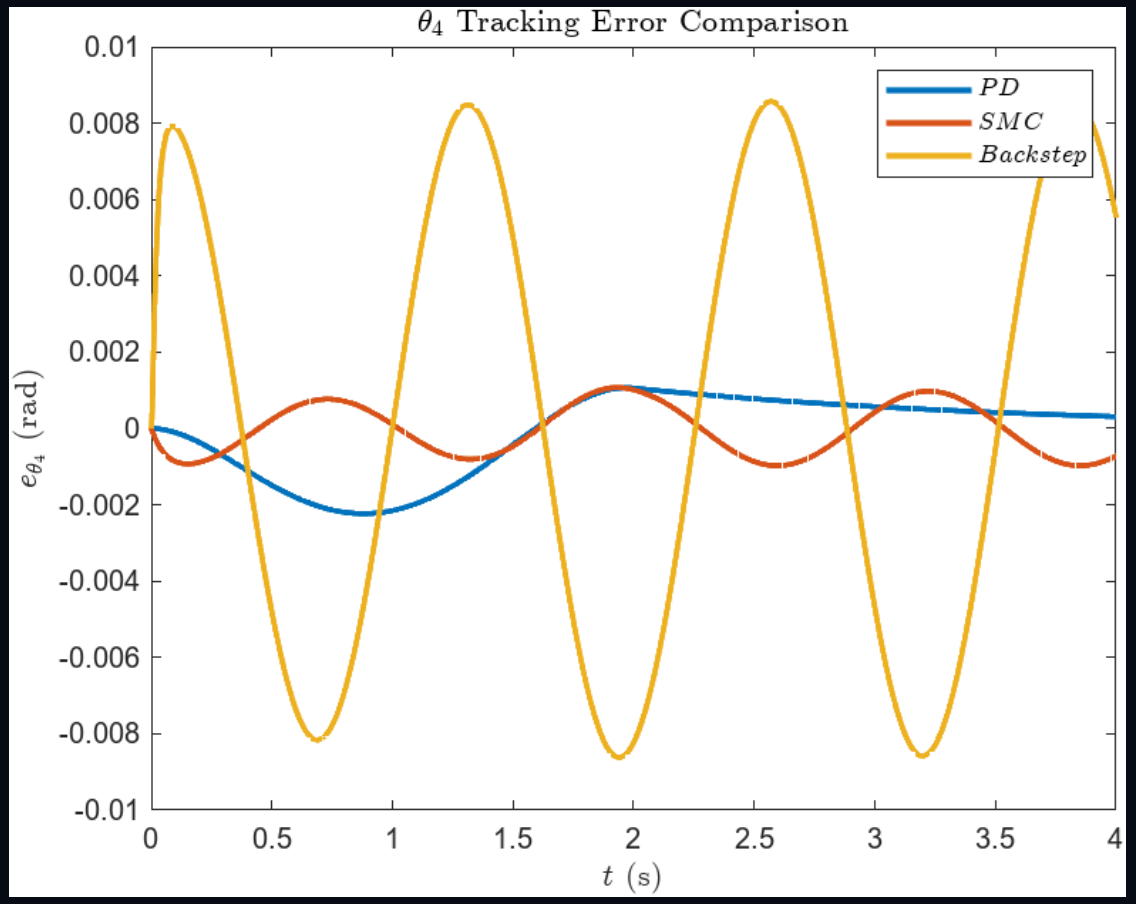
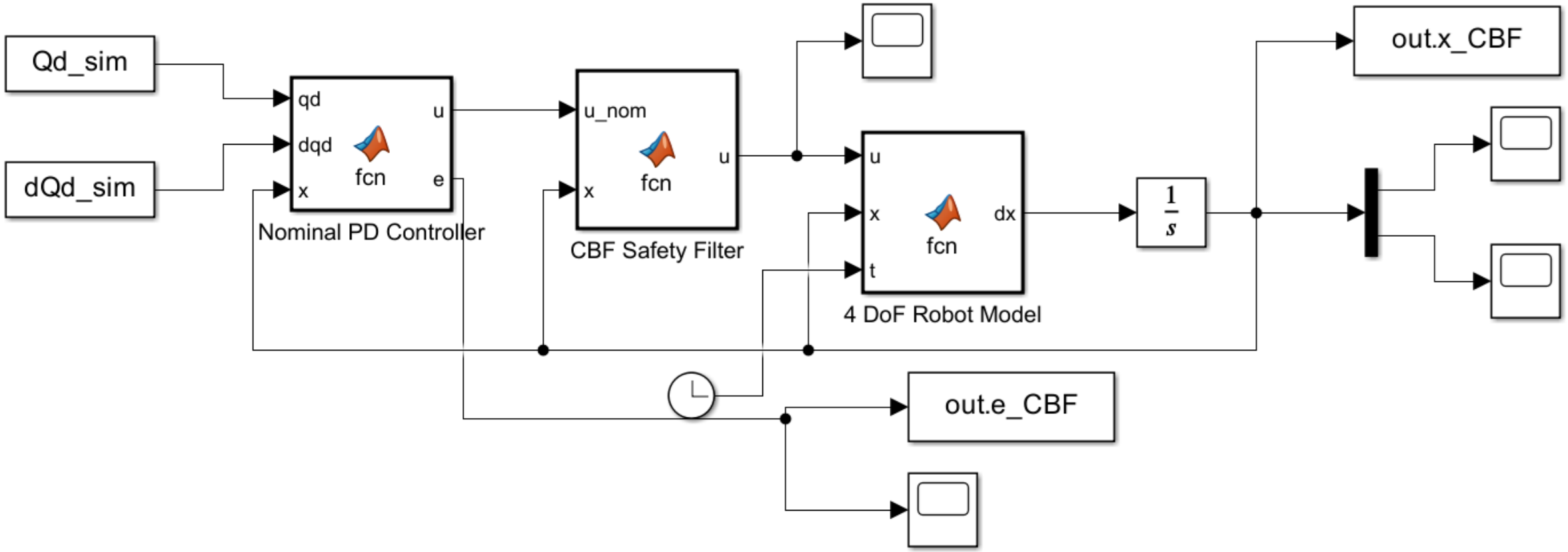




Control Barrier Function Block Diagram



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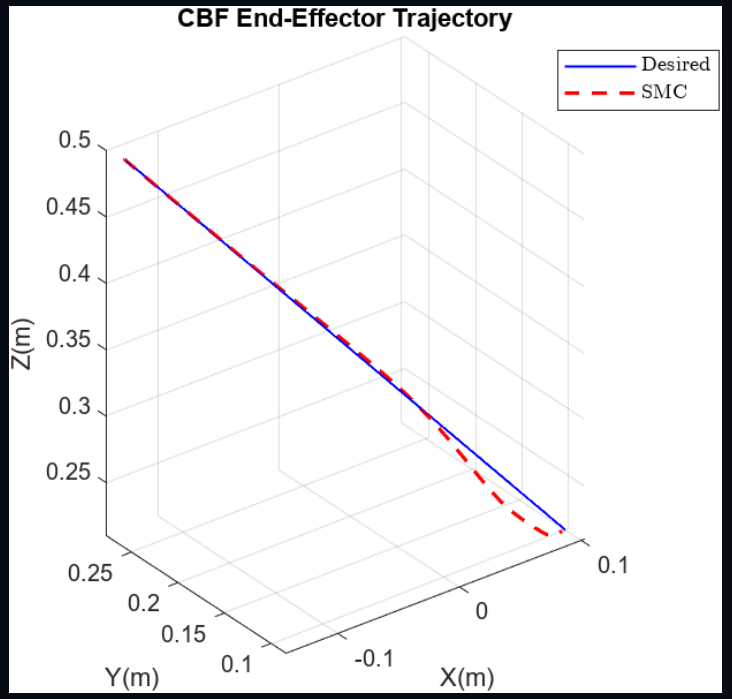
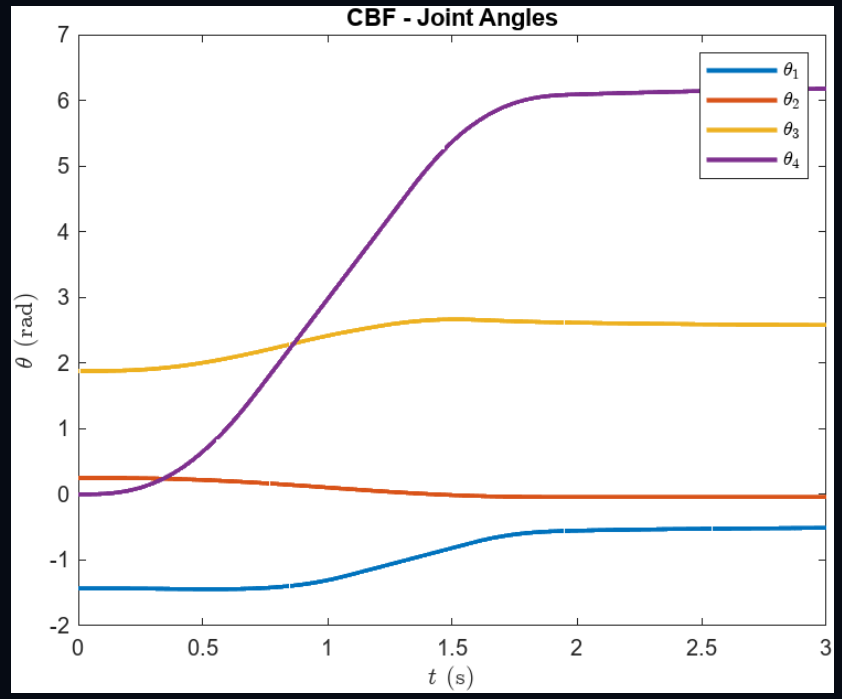
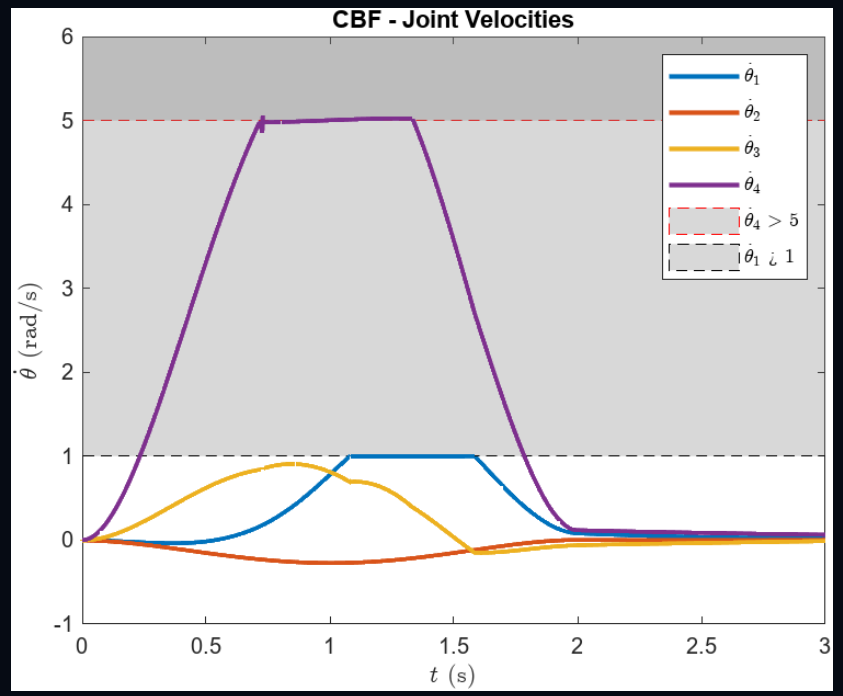
Control Barrier Function

Constraints: Joint Angle 1 velocity < 1 rad/s & Joint Angle 4 velocity < 5 rad/s







References

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[2] S. Alouane, M. Bettayeb, N. Djeghali, R. Fareh, A. Aouaichia, and A. Hamoudi, “Backstepping Active Disturbance Rejection Control for a 4 DoF Robot Manipulator Using High-Order Sliding Mode Extended State Observer,” in *2025 International Conference on Control, Automation and Diagnosis (ICCAD)*, Barcelona, Spain: IEEE, Jul. 2025, pp. 1–6. doi: 10.1109/ICCAD64771.2025.11099273.

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Github Repo

https://github.com/Pmorlando/NonlinearProject4Dof

Questions